

Automation System To Notify SCHOOL PRAYER TIME AND PERIOD-OVER

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This school automation system for reminding the school prayer or national anthem time and period-over bell utilises Arduino to automate the bell schedule and integrate the playing of the national anthem in schools or institutes. The system comprises an Arduino micro-controller, an RTC DS3231 module, a microSD card module, and an audio

jack to connect the speaker.

Fig. 1 illustrates the core functionality of this system, which eliminates the need for manual bell ringing, ensures smooth transitions between periods, and reduces the administrative burden on school staff. The components required for constructing this school automation system are listed in Table 1.

Software

The logical software is written in Arduino IDE. After installing the related libraries, including Wire.h for I2C communication, SD.h for SD card interfacing, and TMRpcm.h for audio playback, upload the code. Before uploading the given sketch to the Arduino Uno board, ensure that the correct board is selected. After selecting the port, upload the sketch file 'code.ino' to the board. Fig. 2 shows a screenshot of the source code.

The provided Arduino program is designed to automate the ringing of a school bell and play school prayer or national anthem at preset times. It uses various libraries and hardware components to achieve this functionality:

1. RTC (real-time clock). Keeps track of the current time, even when the Arduino is powered off. It ensures the system operates on the correct schedule

2. SD card module. Stores audio files, including the prayer or national anthem, to be played through a speaker

3. TMRpcm library. Handles audio playback from the SD card

4. Buzzer. Emits a sound at the start and end of each period to signal the change

Algorithm

1. Initialisation.

- Set up serial communication for debugging
- Initialise the SD card and check for successful initialisation
- Initialise the RTC and set the current time, if needed (commented out in the code)
- Set up the TMRpcm library for audio playback and configure the speaker pin

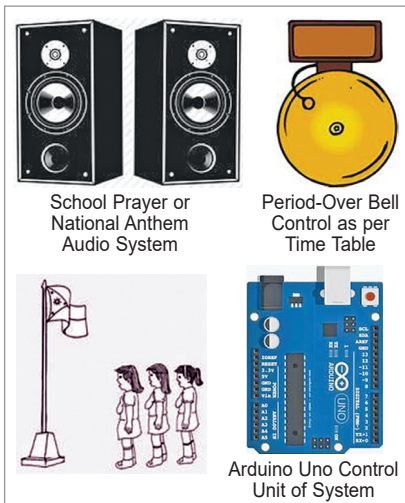


Fig. 1: Essence of this system

TABLE 1 BILL OF MATERIALS		
Components	Quantity	Description
Arduino Uno	1	Control unit of project
RTC 3231	1	Real time clock
Microcard SD module	1	For store audio track
5V battery/power bank	1	5V
3.5mm audio female connector	1	To interface the speaker

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Code
#include <Wire.h> // Include the Wire library for I2C communication
#include <SD.h> // Include the SD library for SD card
#include <TMRpcm.h> // Include the TMRpcm library for audio playback
#include <RTClib.h> // Include the RTClib library for RTC

#define SD_CS_PIN 4 // Pin connected to the CS of SD card module
#define RTC_SDA_PIN A4 // Pin connected to SDA of RTC module
#define RTC_SCL_PIN A5 // Pin connected to SCL of RTC module
#define AUDIO_OUT_PIN 9 // Pin connected to audio jack
#define BUZZER_PIN 10 // Pin connected to the buzzer

RTC_DS3231 rtc; // Create an instance of RTC_DS3231
TMRpcm tmrpcm; // Create an instance of TMRpcm

void setup()
{
    Serial.begin(9600); // Initialize serial communication
    Serial.println("start");
    // Initialize SD card
    if (!SD.begin(SD_CS_PIN))
    {

```

Fig. 2: Snippet of the source code